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Dynamic material flow control in mixed model assembly lines ☆

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ABSTRACT

This study investigates the control of material flow in mixed model assembly lines. It focuses on the use of tugger trains to feed stations in assembly lines by materials and parts from a warehouse or a supermarket. The movement of tugger trains is based on the principle of in-plant milk run. The study considers a strategy to deal with disturbances such as machine breakdown, line stoppage, defective parts, and resequencing of product models. These disturbances lead to unexpected fluctuations in stations demand for parts. The strategy is applied using a mix between the demand-oriented and e-kanban systems to facilitate the planning of three problems, namely, train routing, scheduling, and loading. The information obtained using e-kanban is combined with the information about the expected stations demand based on previously known sequence of product models and the materials needed for each model. Routing was investigated analytically while scheduling and loading problems were investigated using integer programming. Results showed that the method proposed outperforms the traditional methods of material flow planning.

1. Introduction

In mixed model assembly line (MMAL) environment, different product models can be assembled on the same line to fulfill the diversified customer needs. The planning process for MMAL contains several tasks, one of which is the supplying process of materials just-in-time (JIT) from the warehouse or a supermarket. Any delay of materials replenishment can result in stoppage of the assembly line which means idle time for several workers. On the other hand, too early material replenishment can lead to accumulation of materials in the area near the stations which is very scarce and expensive (Baudin, 2004).

A recently used transporter of materials is the tugger train which depends on the milk run principle. This principle is suitable for the repetitive JIT supply of materials and parts in small containers (bins) (Kilic, Durmusoglu, & Baskak, 2012). Every certain period, called *cycle time*, a group of stations are supplied by materials from a warehouse or a supermarket. Supermarkets are scattered inventory areas playing the role of intermediate storage between the main warehouse and the stations for the purpose of being close to these stations (Emde & Boysen, 2012b; Emde, Fliedner, & Boysen, 2012). The group of stations supplied by the same tugger train is called a *cell*. The train cycle time (TCT) is different from the station cycle time (SCT) which is the time interval between two successive product models coming out from the station. To plan this in-plant milk run system, train routing, scheduling, and loading must be determined (Emde & Boysen, 2012b; Emde et al., 2012). In routing problem, the assignment of each train to a group of stations is accomplished. It is found in the practice that sometimes routes are fixed and determined independently from the other two tasks to standardize the system. To facilitate that, certain colors are used on the aisles for certain routes. In scheduling problem, TCT and the beginning of train movement are determined. In loading problem, the exact material loaded in each cycle is determined. The number of bins for each material type must be determined to fulfill stations' demand. Sometimes in peak demand periods, the total demand of stations is higher than the capacity of tugger train. In this case, bins can be loaded before they are actually needed. This type of loading, however, increases the inventory holding cost in a very scarce area.

To plan all these problems, the future demand of stations must be estimated. The exact station demand for materials and parts is usually determined using two systems, namely, demand-oriented and kanban (Klenk, Galka, & Günthner, 2012). In the first one, the exact demand is usually known for the next few shifts depending on the knowledge of workpieces (product models) sequence and the needed materials by each workpiece (Emde & Boysen, 2012b). The second system uses regular kanban or electronic

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kanban. However, on the ground the exact demand is not easily determined because of the disturbances that can happen (Choi & Lee, 2002). Machine breakdowns, workstation starvation, line stoppage, defective parts, and resequencing are just examples for these disturbances. So the planning process depending on the determination of stations demand must be dynamic and flexible. Except the study by Choi and Lee (2002) which considered the case of resequencing, according to the best of the knowledge of the authors, our paper is the first one that concentrates on dealing with all the disturbances mentioned above in the milk run system. The difference between our study and the study by Choi and Lee (2002) is explained in the next section.

In this study, tugger train routing, scheduling, and loading problems are investigated assuming dynamic environment that is subjected to disturbances. To facilitate that, a mix between the demand-oriented and electronic kanban systems is used. This study is organized as follows: Section 2 is about the theoretical background and literature review. Section 3 presents the need for dynamic planning. Section 4 is about routing problem. Section 5 is about scheduling problem. Loading problem is in Section 6. Section 7 presents the results and analysis. Finally, Section 8 presents the conclusion of the study

2. Theoretical background and literature review

To explain the approach in this study some concepts should be clear. In MMAL, a station may be *closed* or *open* at either side of a station boundary. In a closed station, the operator must work within the boundary limits of the station. In an *open* station, an operator is permitted to move outside his station up to some specific limits (Sarker & Pan, 2001). It is possible for stations to be combinations of the two previous types. *Work congestion* occurs when the assembly work is done beyond the station downstream limit, while *work deficiency* occurs when the assembly work is done beyond the station upstream limit (Macaskill, 1973).

Because of the varying assembly time of different models, an operator tends to have an uneven amount of work. As a result, the assembly line is expected to experience *blocking* and *starvation* phenomena. Blocking results in *utility time* (*over load*) which occurs when an operator cannot finish his work on a product model before reaching the workstation boundary downstream and needs additional operator(s) to finish the incomplete product model. *Starvation* results in the *idle time* which occurs when an operator needs to wait for a product model to enter the upstream boundary of his workstation area (Sarker & Pan, 2001). There is another type of workstation starvation which occurs when there is a shortage in the needed materials coming from the warehouse.

Beside starvation and blocking there are other types of disturbances that affect the work of MMAL. Some of these disturbances come from the failure of some machines, different processing times for different product models, and fluctuations and changes in customer demand. If unforeseen disturbances like machine breakdowns or material shortages occur, a resequencing of a given production sequence often becomes essential (Boysen, Scholl, & Wopperer, 2012). Moreover, in an assembly line of a JIT production system, workers have the power and the responsibility to stop the line when they fail to complete their operations within their work zones (Xiaobo & Ohno, 2000). So, line stoppage and utility work are two alternatives to deal with the case in which the worker cannot finish his/her work on time.

The part feed problem has been widely studied in the literature. For example, Caputo and Pelagagge (2011) investigated the part feeding policy where their analysis focused on three basic policies, namely kitting, just in time kanban-based continuous supply and line storage. Some guidelines and decision tools were provided to assist production managers in the selection of proper policies for components delivery to assembly systems. The research methodology was based on developing theory-based quantitative planning models associated with an empirical approach to assign candidate delivery methods to specific components groups. The forklift system was investigated by several studies. For example, considering Assembly to Order (ATO) or Make to Order (MTO) mixed model assembly lines systems, Battini, Faccio, Persona, and Sgarbossa (2009) investigated both the inventory centralization/decentralization problem and the line feeding policy in a general framework. They considered three feeding policies which are pallet to workstation, trolley to workstation, and kit to assembly line. So they concentrated on three different transport devices (fork-lift, hand truck or trolley, assembly kit). They ignored investigating the use of tugger trains in their feeding policy. A multi-factorial analysis has been carried out and an industrial application of the introduced framework was illustrated. Boysen and Bock (2011) considered the feeding of parts using forklifts from a central warehouse. They presented a case study for a company manufacturing cars. In this company the mixed model assembly line is located in the second floor. So a hoist system is used to get the parts to the second floor where forklifts are ready to take parts and deliver them to the workstations. Part distribution by the forklift is a bottleneck. Therefore, a suitable sequence of given material boxes has to be found. Furthermore, a direct comparison with a simple two-bin kanban system was provided.

In milk run system, the literature shows that there are four main decisions which are:

- Supermarket location problem (Alnahhal & Noche, 2015; Emde & Boysen, 2012a)
- Tugger train routing (Emde & Boysen, 2012a; Golz et al., 2012; Emde & Boysen, 2012b)
- Tugger train scheduling (Emde & Boysen, 2012b; Golz et al., 2012; Choi & Lee, 2002)
- Tugger train loading (Choi & Lee, 2002; Golz et al., 2012; Emde et al., 2012).

In our study, the last three items which are routing, scheduling and loading problems are considered in the environment of mixed model assembly lines taking into consideration the occurrence of disturbances.

Other types of classifications can be as follow:

- Transport devices: forklift, hand truck or trolley, assembly kit, and tugger train
- Material flow control: demand-oriented and kanban (e-kanban and traditional kanban)
- Production environment: mixed model assembly line and others (such as job shops)
- Inventory type: centralized warehouse and decentralized supermarket

More classifications can be found in Klenk et al. (2012). In this study, tugger trains are used. The study combines, for the first time, the demand oriented and e-kanban types in the same time. In production environment type, mixed model assembly line is considered. Moreover, the two types of inventory can be considered.

Before get into details, a big picture of the decisions should be considered. For example, Battini, Gamberi, Persona, and Sgarbossa (2015) presented an approach for the part feeding for assembly systems in which supermarket storage is adopted and coupled with an automated transportation system. They made a conceptual framework for the design of the system. The supermarket type can be fish bone, single line, or multi line. The transportation mode can be shuttle/conveyor, AGV/LGV, or tow train. The line side presentation can be load unit, station kit, and traveling kit. A decision support matrix was presented where the tow train, AGV, and shuttles are compared based on certain criteria. In most of the cases the tow train was found to be the best option. The transportation system choice was found to be strongly affected by four key parameters: the number of meters traveled during the feeding process, the assembly line dimension (i.e. the number of stations), the assembly line takt time and the number of part bins demanded by stations per takt.

In the literature of milk run system, the control system of material supply is demand-oriented system or kanban system. Several studies concentrated on the first one. For example, Emde and Boysen (2012b) studied the train routing and scheduling together in parallel to decrease the inventory holding costs. Emde et al. (2012) studied the loading problem to minimize the inventory holding cost. The two studies assumed that the feeding process starts from decentralized supermarkets. Golz, Gujjula, Günther, Rinderer, and Ziegler (2012) studies the routing, scheduling, and loading problems in a centralized inventory system. Outside the milk run system, train scheduling is known in the literature. For example, Babu et al. (2014) studied train scheduling, ship scheduling and stockyard management in the context of coal imports in port terminals.

One the other hand, other studies concentrated on the kanban system. Bozer and Ciemnoczolowski (2013) and Ciemnoczolowski and Bozer (2013) investigated the traditional kanban system where they determined the number of kanban assuming a given workstation starvation level. Before that, they presented the distribution of demand for bins assuming that the demand is always one or two bins for each workstation during the train cycle. A study by Faccio, Gamberi, Persona, Regattieri, and Sgarbossa (2013b) used simulation to analyze the system. In another study by Faccio, Gamberi, and Persona (2013a), a cost model has been proposed using analytical approach. They concentrated on the special attributes of such a system and they provided an innovative procedure to optimally set all decision variables. A study by Faccio (2014) also investigated the milk run kanban system and compared it to kitting system in the case of using decentralized supermarket. The comparison was based on the impact of production mix variability and of the models variety. Bortolini, Ferrari, Gamberi, Manzini, and Regattieri (2015) presented a model for feeding the parts from supermarket to mixed model assembly lines and presented a real case study. The kanban system is used to regulate the material flow. There are three costs drivers which are material handling cost, inventory cost of the assembly lines, and stock out cost. The tradeoff between the fleet size and the size of line side inventory has been addressed. Two free decisional variables are considered looking for their best value definition, i.e. the number of carts per tow-train and the so-called security factor.

Dynamic planning for material replenishment was investigated by Choi and Lee (2002) who studied the feeding process in an automotive plant. At first, they investigated the loading problem in which the parts to be fed and the feeding amounts are indentified. Then the feeding orders to feeders are assigned and routes are formed. This is based on the objective of minimizing the deviation of actual delivery time from the ideal one, where a certain safety stock level should be kept. However, the loading problem in the study by Choi and Lee (2002) did not take into consideration peak demand periods in which the shuttles capacities are exceeded, so their study does not contain *early loading* in which some bins are loaded before they are needed. Furthermore, they only considered the case of resequencing. According to the best of the knowledge of the authors, the study by Choi and Lee (2002) was the only one which concentrated on dynamic planning of the feeding system. Our study investigates this dynamic planning process in terms of routing, scheduling, and loading problems and considers the peak demand periods. We present a system that is able to deal with quality problems, line stoppage caused by machine breakdown or inability of worker to finish his work on time, and resequencing based on disturbances or customer needs.

3. The need for a dynamic system

At first we investigate routing problem. We assume that the routing problem is independent from disturbances and resequencing. The routing problem depends on the total average demand which is usually stable even after resequencing. Stable routing will give some stability in the system. On the other hand, scheduling and loading problems should by dynamic to respond to the changes in the detailed demand of stations. However, in the routing problem in this study, the effect of early loading will be taken into consideration. The results of the scheduling problem are used as inputs for loading problem. However, scheduling was investigated in a way that guarantees the feasibility of loading problem regarding the capacity of the train. Scheduling and loading problems in our study take into consideration the current material amounts in line-side inventory after each disturbance. The objective is to make these amounts of materials as close as possible to the normal safety stock level as fast as possible. The objective is done considering minimizing the number of cycles, the probability of early loading, and the deviation of actual line-side inventory from the ideal safety stock level.

The following basic assumptions are made in this study:

- There is no any disturbance in the movement of the tugger trains. The disturbances only occur in MMAL
- The demand of stations can be represented by Poisson distribution
- If there is no disturbances for a certain period of the shift, the stations' exact demand for parts can be expected in this period.
- The time between two disturbances is long enough to fix the problems caused by the first disturbance. These problems are basically the deviation of the size of line-side inventory from the ideal safety stock size.

Basically the material control system can be categorized as demand-oriented or kanban. In demand-oriented system, the exact demand is assumed to be known in at least the next shift. This fact makes the demand oriented-system able to deal with peaks period of demand using early loading. However, there are some cases that make it difficult to anticipate the exact stations demand for parts. Some of these cases are:

1. Resequencing: as stated before, the sequence of the product models can be changed during the shift based on changes in demand or some disturbances. In this case the total demand during the shift remains the same but deviates from the original plan over time.
2. Disturbances: for example in the case of line stoppage, the demand can be shifted by stoppage time. Some minor changes in demand can occur in the case of utility work. In this case the demand can be delayed. Another problem arises when the next station in which utility work is done has the same type of material of the previous station. In this case the utility worker may use the material of this next station. So the exact quantity of material in the two stations' line-side inventory is no longer known. The same problem can occur in the case of work deficiency.
3. Quality problems: if there are some defective parts, some extra parts must be provided to compensate for the defective ones.
4. Unpaced MMAL: in an *unpaced line*, the tasks of the workstations can be decoupled from each other. So, each of the workstations operates independently (Smunt & Perkins, 1985). In

this system, in-process inventory is needed among stations, where it is a design factor (Chakravarty & Shtub, 1985). In this case the exact demand times cannot be known.

This study does not cover the last case. In the other cases, the kanban system can be used since it depends on the continuous feedback sent to the supermarket area about the status of the parts consumptions. To enhance the performance of the system, e-kanban can be used to send the signals instantly. However, this study is about getting the advantages of both the e-kanban and the demand-oriented systems to design a dynamic planning system. In this system, the expected demand during the next few train cycles after disturbances is determined. This is done depending on the information gathered from the signals sent by e-kanban system to the supermarket area about the parts consumption and resequencing details. This is important to know the demand in the next train cycles and to exactly know the amount of parts in line-side inventory. The exact parts consumption can be obtained using radio frequency identification (RFID) system or bar code. Traditional kanban system is not very effective in the proposed approach in this study because there are some delays in sending the information about the exact size of consumption of each station.

The following parameters must be known:

dp_i: number of defective parts of type i during the previous disturbance period
d_{ik}: the expected number of parts i demanded in station cycle k according to the new workpieces sequence.
CP_i: the number of consumed parts of type i during the previous disturbance period
PD_i: the planned number of delivered parts of type i during the previous disturbance period
β_i: ideal safety stock size for parts i

Then current line-side inventory for parts $i(\alpha_i)$ can be estimated as follows:

$$\alpha_i = \beta_i - (\mathrm{CP}_i - \mathrm{PD}_i + dp_i) \tag{1}$$

This value will be used later as input in the scheduling and loading problems. The disturbance period is the time interval at which $dp_i > 0$ or $\mathrm{CP}_i \neq \mathrm{PD}_i$. The RFID or bar code is important to detect dp_i and CP_i values. The previous five parameters can be converted to bin size units if the detection of each part is not possible. In this case, the bar code or RFID tag can be attached to each bin. In this study, we assume that there is no any disturbance for the delivery of the train. The disturbances can only occur in the assembly line. This means that PD_i is not affected by disturbances. In the case that early loading is not used, the planned number of delivered parts is equal to the expected demand in previous train cycles. Another possibility for that equivalence is that disturbances occur in bottleneck periods of demand.

In the case of disturbances, the term $\mathrm{CP}_i - \mathrm{PD}_i + dp_i$ in the previous equation can be with a positive or a negative sign depending on the following factors:

- The possibility of using early loading: if early loading is used the term tends to be negative
- The time of disturbances: if they happen just after peak-demand period, the extra material resulted from early loading is consumed. Therefore, the sign in this case is based on the nature of disturbances.
- The nature of disturbances: defective parts make it positive while delays caused by line stoppage or machine breakdown make it negative.

In the case of machine breakdown in a station, it is expected that the stations before that station have consumed parts more than the stations after it. Therefore it is expected that the first group of stations have lower line-side inventory. So its probability for starvation increases. In the following three sections, routing, scheduling, and loading problems, specially designed to be suitable for dynamic planning, are presented.

4. Fixed routing and effect of early loading

In the case of dynamic material flow control, there are two main advantages for fixed routing:

1. Making planning easier: after resequencing, for example, some tugger trains are in the supermarket area while others are still in their routes. If variable routing is to be done, a plan must take into consideration these two groups of trains and assign each train to the right stations. This complicates the planning process.
2. Providing some extra train capacity that is useful to absorb the effect of sudden increases in demand in some cycles.

Although the stations demand is very dynamic, in the practice sometimes the routes are fixed, where certain colors are used to determine certain routes. To maintain a certain probability of *capacity fill rate* (δ), the maximum permitted average demand of stations in the cell must be determined. Capacity fill rate, or simply *fill rate*, in this study refers to the probability of providing the cell with all the needed bins. In this study, fill rate depends only on the capacity of the train. The fill rate for all the cycles in the shift can be obtained by multiplying the fill rate for each cycle. Low fill rate causes high level of workstation starvation.

So if early loading is not used, it is possible to write

$$\delta_i = \min\left(\frac{K}{D_i}, 1\right) \tag{2}$$

$$\delta = \prod_{i=1}^{T} \delta_i \tag{3}$$

where K is the capacity of the tugger train feeding the cell, T is the number of train cycles in the shift, and D_i is the cell demand in the train cycle i. The cycle demand for each station in the cell is for the time interval from the arrival of the train at the station to the next train arrival at the same station in the next train cycle. Later the fill rate is investigated if early loading is used. The average demand can be controlled by determining the number of stations in the cell. But before that we find the fill rate if the average demand is exactly as the capacity of the train. In the calculations, we assume that early loading will be done later.

Decreasing fill rate increases stations starvation. This starvation is different from another type of starvation which occurs when an operator needs to wait for a workpiece to enter the upstream boundary of his workstation area. Generally, the station starvation is caused by the following points:

1. The total accumulated demand of all stations in the cell must always be less than or equal to the capacity of the train multiplied by the number of train cycles.
2. In the case of early loading, for each station the maximum line-side inventory (MLSI) must not exceed the capacity of the area beside the station.
3. In the case of traditional kanban system, the number of kanban affects the starvation probability

The previous first point is related to the fill rate. If any of the two constraints in the first two points were not realized, workstation starvation occurs. The effect of MLSI can be negligible if the number of stations is high. This is because there are a lot of chances to make early loading without exceeding MLSI.

Suppose that the average cell demand per cycle is equal to the capacity of the train ($D = K$), and assume that the demand of stations for bins can be represented by Poisson distribution. $\delta_{\mathrm{EL2}}(i)$ represents the fill rate for the current and previous routes (i and $i-1$) if early loading is used. It can be obtained using Eqs. (4) and (5)

$$\delta_{\mathrm{EL2}}(i) = \sum_{j=0}^{D(i-1)} \mathrm{pmf}(D(i-1)-j, D(i-1))\mathrm{CDF}(D+j, D), \quad D=K \quad \forall i = 2\ldots T \tag{4}$$

In Eq. (4), if $i = 1$, there is no previous route, so $\delta_{\mathrm{EL2}}(1)$ is determined using Eq. (5)

$$\delta_{\mathrm{EL2}}(1) = \mathrm{CDF}(D, D), \quad D = K \tag{5}$$

where

$\mathrm{pmf}(k, \lambda) = \Pr(X = k)$ is probability mass function

$\mathrm{CDF}(k, \lambda) = \Pr(X \leqslant k)$ is cumulative distribution function

In Eq. (4), the term "$D(i-1)$" represents the total average cell demand for bins in all the cycles of the train before the current 'i' cycle. Because $D = K$, the term also represents the total capacity of the train cycles before the current one. So if the total actual demand in the train cycles before the current cycle is "$D(i-1)-j$", then the previous cycle will be feasible (without starvation). For the current train cycle to be feasible, the demand for parts in it must be at maximum "$D + j$". That means that any savings represented by 'j' in the bins demand (consumption) in the previous train cycle will be advantageous for the current train cycle because of the early loading principle. The probability of that can be estimated using the second term in the right side of the equation which is "$\mathrm{CDF}(D + j, D)$". All the possible values of 'j' are taken into consideration. So the equation is to estimate the probability that both the current and the previous train cycle are feasible, and this feasibility is called fill rate.

It was found using simulation that $\delta_{\mathrm{EL}}(i)$ which is the fill rate for routes from 1 to i can be estimated as in Eq. (6)

$$\delta_{\mathrm{EL}}(i) = \delta_{\mathrm{EL2}}(1) i^{-\delta_{\mathrm{EL2}}(i)} \quad D = K \tag{6}$$

So the fill rate for all the routes is

$$\delta_{\mathrm{EL}}(T) = \delta_{\mathrm{EL2}}(1) T^{-\delta_{\mathrm{EL2}}(T)} \quad D = K \tag{7}$$

It was found using simulation that the effect of the number of routes is very small when fill rate is high. So fill rate is almost independent from the number of cycles. This fact makes calculations easier. Using simulation, it was found when fill rate is high that:

$$\delta_{\mathrm{EL}} \cong \mathrm{CDF}(K, D) \quad K \gg D \tag{8}$$

That means that the fill rate depends only on the capacity of the train and the average cell demand. Later in this paper, Eq. (10) will be obtained from Eq. (8) to find the number of stations in the cell for a desired level of fill rate. Fig. 1 shows how the values obtained using simulation and the values obtained using Eq. (8) approaches each other when the fill rate is high. The two lines are almost the same in the right side of the figure. The data used to obtain this figure was the following: there are 10 workstations each of which has an average demand of 2 bins per train cycle, and there are 10 train cycles in the shift. The demand is assumed to be Poisson distributed for all the stations. To obtain the simulation line, 20,000 randomly generated instances were used, where the average value for the fill rate is used in Fig. 1. Eq. (9) was used to check the feasibility of a given instance containing the detailed demand of a group of workstations.

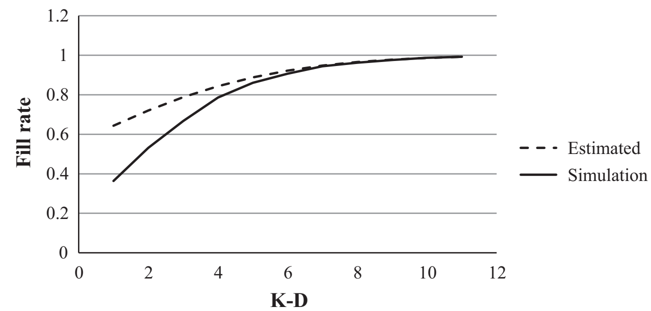


Fig. 1. The difference between the results of Eq. (8) and the results of simulation.

$$\sum_{s=1}^{N} d_{st} \leqslant tK \quad \forall t = 1\ldots T \tag{9}$$

where d_{st} is the demand of station s for bins in the train cycle t.

The probability of feasibility based on Eq. (9) is the fill rate. Increasing the number of train cycles will decrease the fill rate. The 'Estimated' line is obtained using Eq. (8). The data previously mentioned was altered several times and in each time the same behavior shown in Fig. 1 occurs.

Since Eq. (8) ignores the effect of the number of train cycles, using simulation Table 1 shows the difference between the obtained fill rate assuming two cases. In the first one, the number of train cycles equals 1, and in the second one the number of train cycles equals 10. This difference is obtained by subtracting the two values in the two cases and dividing the result by the value in the second case. The effect of the number of the train cycles decreases when the fill rate increases. Fig. 1 and Table 1 show the accuracy of Eq. (8).

So to estimate the number of stations (N) supplied by the same train, we use

$$K \geqslant F^{-1}\left(\delta_{\mathrm{EL}}, \sum_{i=1}^{N} d_i\right) \quad K \gg D \tag{10}$$

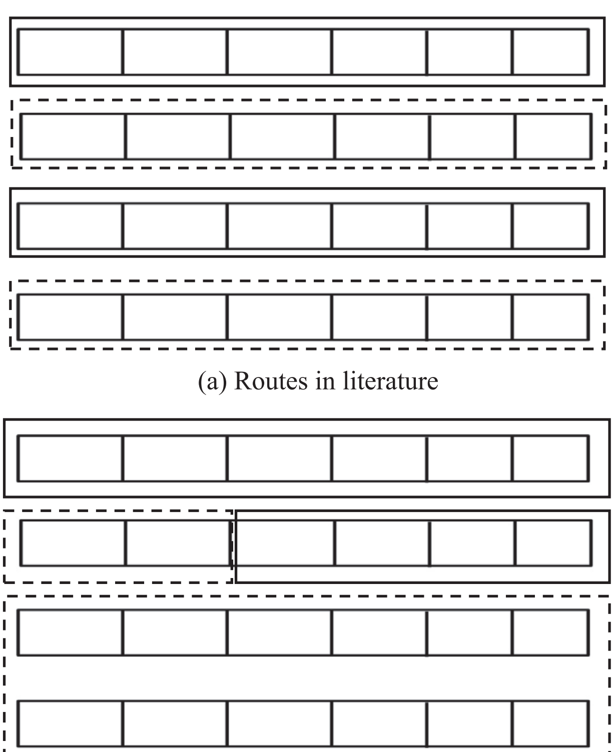
where d_i is the average cycle demand for station i, and F^{-1} is the CDF inverse. The maximum value of N without breaking the previous constraint is found. N is the maximum value that does not break constraint (10). The transportation and loading/unloading times must be considered in the determination of TCT. The summation of these times for all the stations increases when the number of stations increases. This means that TCT must be increased and hence d_i increases. So the values of d_i are recalculated every time the number of stations is changed. The demand can be based on the minimum

Table 1
Effect of number of train cycles per fill rate.

$K - D$	Fill rate when there are 10 train cycles (%)	Difference between fill rates when there are 1 and 10 train cycles (%)
1	36.38	75.42
2	53.17	35.02
3	66.76	16.83
4	78.60	7.53
5	86.04	3.32
6	90.64	1.66
7	94.47	0.71
8	96.20	0.31
9	97.63	0.11
10	98.71	0.03
11	99.25	0.02

possible TCT for the given number of stations. This is because the priority is to decrease the number of tugger trains as much as possible. This means that in the peak periods of demand, TCT must be the minimum possible one to be able to provide the needed cell demand using the maximum possible capacity without waiting in the supermarket area.

The cell can be supplied by a central supermarket or by decentralized supermarkets. One decentralized supermarket may feed several assembly lines. However, sometimes one decentralized supermarket supplies just part of the assembly line (Emde & Boysen, 2012b; Emde et al., 2012). In the case of using a central supermarket or a decentralized supermarket feeding more than one assembly line, we use a different approach from those found in the literature. The serpentine routes method used in Golz et al. (2012) is not very effective in the case of using fixed routes since it results in high level of underutilization of tugger trains. Its philosophy is that the train must feed the whole assembly line and not just a group of stations inside it for many advantages stated in Golz et al. (2012). Fig. 2 compares the serpentine routes method and our method of routing. The stations in the solid line belong to the same cell. The stations in the dotted line belong to another cell. In the first method, the first and third assembly lines are supplied by the same train in the same train cycle, while the second and third assembly lines are supplied by another train in another. On the other hand in our method, two stations of the second assembly line are supplied by a train while the rest of stations are supplied by another train. So in this study the assembly line can be supplied by more than one tugger train. This is more appropriate in the case of fixed routing. This method is more flexible and enhances the utilization of the tugger train but can lead to traffic jam of the trains if there is more than one train in a narrow aisle. In the case of using a decentralized supermarket feeding just part of the assembly line, routing is simpler. In the two cases, the routing is as follows: just start from the first station and go until the final possible station while keeping the constraint in Eq. (10) in mind. When reaching the last station in the cell, begin a new cell starting from the station after it and continue in the same way.



(a) Routes in literature

(b) Routes in this study

Fig. 2. Two ways for routing.

5. Scheduling

In the period just after disturbances, there are two main special factors for the scheduling process:

1. It must consider the size of the current line-side inventory. MLSI resulted from scheduling plus the line-side inventory remaining from previous period must not exceed the maximum capacity of the space near the stations.
2. More attention is paid for the capacity of tugger trains, where sometimes, the train capacity is still not enough even after using early loading. So scheduling process must take into consideration that the loading problem that will be conducted later is feasible regarding the accumulated stations demand. So high fill rate should be guaranteed.

The first factor is also in loading problem. The objective function of scheduling is minimizing the number of train cycles and possibility of early loading. The maximum train capacity and the maximum capacity of the space near the stations must be taken into consideration. Minimizing the number of cycles reduces the efforts and the traffic jam in the facility, while minimizing the probability of early loading reduces the inventory holding costs.

The following parameters and variable are used:

MC is the minimum possible TCT minus 1 (represented in SCT unit).

TSC is the total number of station cycles in the shift

$$x_{ij} = \begin{cases} 1, & \text{if there is a train cycle that covers the stations cycles from } i \text{ to } j \\ 0, & \text{otherwise} \end{cases}$$

d_{ku} is the demand (in bins unit) of station k during the station cycle u

M is the maximum capacity of the area beside stations

w_i is the weight of the term i in the objective function

The minimum possible TCT is based on the total needed time for train transportation and loading and unloading of bins beside the stations and in the supermarket area. In determining MC, we used "minus one" to make the equations simpler. MC represents, in station cycle time unit, the difference between the first and the last station cycles covered by one train cycle. For the estimation of d_{ku}, for a group of stations belonging to the same assembly line if the station cycle u for a station k is τ, then the associated u for the station $k+1$ is $\tau+1$. This is because it takes one SCT for a workpiece to move from the station k to the station $k+1$. Table 1 shows the associated u values for different stations and cycles. Any column in the table represents the cycles in which the demand of the same workpiece take places on different stations. As it is obvious, even when $u=\tau+3$ for the fourth station, d_{ku} is written not as $d_{k(\tau+3)}$ but as $d_{k\tau}$. In other words, the station cycle number depends not only on the time of the cycle but also on the station number (see Table 2).

Table 2
Stations demand representation method.

k (station number)	Station cycles		
	$u=\tau$	$u=\tau+1$	$u=\tau+2$
1	τ	$\tau+1$	$\tau+2$
2	$\tau+1$	$\tau+2$	$\tau+3$
3	$\tau+2$	$\tau+3$	$\tau+4$
4	$\tau+3$	$\tau+4$	$\tau+5$
d_{ku}	$d_{k\tau}$	$d_{k(\tau+1)}$	$d_{k(\tau+2)}$

The model is as follows:

Objective function

$$\min w_1 \sum_{i=1}^{TSC-MC} \sum_{j=i+MC}^{TSC} y_{ij}^{+} + w_2 \sum_{i=1}^{TSC-MC} \sum_{j=i+MC}^{TSC} x_{ij} + w_3 \sum_{i=1}^{TSC-MC} \sum_{j=i+MC}^{TSC} x_{ij} \sum_{k=1}^{N} \sum_{u=i}^{j} d_{ku} \quad (11)$$

Subject To

$$x_{ij} \sum_{u=i}^{j} d_{ku} \leqslant M \quad \forall i = 1, \ldots, \mathrm{TSC}-\mathrm{MC}, \quad \forall j = i+\mathrm{MC}, \ldots, \mathrm{TSC}, \quad \forall k = 1, \ldots, N \quad (12)$$

$$x_{ij} \sum_{k=1}^{N} \sum_{u=i}^{j} d_{ku} - y_{ij}^{+} + y_{ij}^{-} = K \quad \forall i = 1 \ldots \mathrm{TSC}-\mathrm{MC}, \ \forall j = i+\mathrm{MC} \ldots \mathrm{TSC} \quad (13)$$

$$\sum_{i=1}^{I-\mathrm{MC}} \sum_{j=i+\mathrm{MC}}^{I} x_{ij} \sum_{k=1}^{N} \sum_{u=i}^{j} d_{ku} - \sum_{k=1}^{N} \alpha_k \leqslant K \sum_{i=1}^{I-\mathrm{MC}} \sum_{j=i+\mathrm{MC}}^{I} x_{ij} \quad \forall I = \mathrm{MC}+1 \ldots \mathrm{TSC} \quad (14)$$

$$\sum_{i=1}^{j-MC} x_{ij} = \sum_{v=j+1+\mathrm{MC}}^{\mathrm{TSC}} x_{j+1,v} \quad \forall j = \mathrm{MC}+1, \ldots, \mathrm{TSC}-\mathrm{MC} \quad (15)$$

$$\sum_{j=1+\mathrm{MC}}^{\mathrm{TSC}} x_{1j} = 1 \quad (16)$$

$$\sum_{i=1}^{\mathrm{TSC}-\mathrm{MC}} x_{i\mathrm{TSC}} = 1 \quad (17)$$

$$x_{ij} = 0 \ or \ 1, \quad y_{ij}^{+}, y_{ij}^{-} \geqslant 0 \quad \forall i = 1, \ldots, \mathrm{TSC}-\mathrm{MC}, \ \forall j = i+\mathrm{MC}, \ldots, \mathrm{TSC} \quad (18)$$

Appropriate weights for the objective function terms can be determined by management staff according to the importance of minimizing inventory holding costs and the number of cycles. The first term in the objective function is to minimize the possibility of early loading in loading problem. This objective is important to minimize the inventory holding cost because early loading increases the average line-side inventory. The second term is to decrease the number of train cycles. The third term is to guarantee that there is no overlap in the cycles. Constraint (12) is to guarantee that demand does not exceed the capacity of area beside stations. Constraint (13) is to define the variable y^{+}_{ij} used to minimize the possibility of early loading. When y^{+}_{ij} is larger than zero, it means that the demand is more than the capacity for certain train cycle, and that means the necessity for early loading. In the objective function y^{+}_{ij} is put because the objective is reduce it. On the other hand, if y^{-}_{ij} is larger than zero, it means that the capacity is more than the demand. Constraint (14) is to prepare for early loading in the next section where it guarantees that early loading will be feasible regarding the accumulated demand constraint. The current line-side inventory for parts k (α_k) computed in Eq. (1) has been taken into consideration. In the case that there are some train capacity problems (not very high fill rate), constraint (14) will make the model infeasible. So it can be rewritten in the same way of constraint (13), and that is using goal programming to make the constraint soft. Constraint (15) is to guarantee that all the station cycles demand is fulfilled. If a certain train cycle for the station cycles i to j is chosen in the solution, there must be a new train cycle starting from the station cycle $j + 1$. Together with constrains (16) and (17), the fulfilling of cycle demands are guaranteed by using constraint (15). Constraints (16) and (17) are to guarantee that there is an initial train cycle covering the first station cycle, and a final train cycle to cover the last station cycle demand. Without these two constraints, all variables x_{ij} will be zeroes.

To show the effect of the weight of each term in the objective function, a simple example is presented in Table 3, where the 1's represent the demand for bins in the station cycles. For the first station, there was no any demand until the 17th station cycle because the station has some parts from the previous shift. Only the station cycles in which the demand occurs were shown. In the real life, the numbers of station cycles and stations are usually larger. In the example, it was assumed that MC = 8, M = 4, and K = 6. Moreover, α_k = 2 for all the stations.

The third term in the objective function can be multiplied by a very small number because it is only to guarantee that there is no extra number of train cycles. For the first and second terms, the results depend on the value that is multiplied by each term, where the results are as in Table 4.

As in the first scenario if the weight is zero, there is no penalty if the demand is more than the capacity of the train. In this case the penalty will only be for the number of the cycles of the train. Therefore, there are only two cycles, and the demand for the train is more than its capacity by 1 bin in the first cycle and 6 bins in the second cycle. Because α_k = 2 for all the stations, the solution is still feasible, but the line side inventory size will be deviated from its initial one. In the second scenario the weight for the first term was set to be 1, and there are three cycles. There is still one cycle with demand more than K. In the third scenario, the importance of the first term was set to be very high on the expense of the number of train cycles, where there are 4 train cycles with no any cycle with demand more than the capacity of train. The weights of the first two terms of the objective function are left for the decision maker based on his/her judgment. Results were obtained using free software called LP solve.

The scheduling and loading problems found in the literature assume that there is no any disturbance. Moreover, the scheduling problem in this study prepares for early loading considering the current size of line side inventory. It guarantees that the loading problem is feasible, where the scheduling problem is an input for the loading problem. Ignoring this fact can lead to infeasible loading problem which leads to workstation starvation. Without using the signals about the real consumption of materials, a lot of problems can happen depending on the type and level of disturbance. Rescheduling may not be problematic because it is a major change that can be easily known by the material handlers using simpler methods of information sharing. However, the effect of some problems in the quality of some parts can be gradually increased and not be known or estimated by the material handlers except if sever workstation starvation happens. The effect of machine breakdowns or line stoppage for short times can be detected only if the line side area is full with bins. It is important for the signals to be electronically sent for fast response by the material handling system. If the current size of line side inventory is not known because the electronic signals are not used and if it is higher than the ideal safety stock level, then the real needed capacity of the tugger train will be less than the assumed one. This will not make a problem in scheduling but must be considered in routing problem. However, the problem will occur if the size of the current line side inventory is lower than the ideal safety stock assuming that the scheduling problem takes into consideration the safety stock level. This is because the real needed capacity will be higher than the assumed one.

In the case that the electronic signals are not used, the planner has two strategies. The first one is to assume that the line side inventory is zero. Usually, the safety stock should not be used except if it is very necessary in some bottleneck periods where the shortage of material will occur. Usually to cover such a

Table 3
Stations demand for bins during the station cycles.

Station	Station cycles												
	1	11	12	14	17	18	21	25	33	34	35	38	40
1					1		1						
2	1	1				1						1	1
3	1			1					1				
4	1	1				1				1			
5	1		1					1	1		1		

Table 4
Results for different weights of the first term in the objective function.

Weight	Train cycle number			
	1	2	3	4
0	7 (1–12)[a]	12 (13–40)		
1	7 (1–12)	6 (13–25)	6 (26–40)	
10	6 (1–11)	5 (12–20)	2 (21–30)	6 (31–40)

[a] $D(i-j)$ means that the train cycle covers the station cycles from 'i' until 'j' and the demand is 'D' bins in this cycle.

shortage, some emergency tugger trains routes are used. The problem for these tugger trains is that if the real line side inventory is not zero is that they add cost which is not necessary and may result in excessive material accumulation beside the stations. The second strategy is to assume that the current line side inventory is equal to the ideal safety stock level. Suppose that in normal situations that the system suffers from 10% probability of workstations starvation if the safety stock is needed. If that system does not use the electronic signals to detect the depletion of all the safety stock, then the system will suffer from 10% probability of workstations starvation.

6. Loading problem

The objective function in this problem is minimizing the deviation between the line-side inventory and ideal safety stock levels. The model will minimize the total inventory holding costs. In the case that sudden large demand quantities higher than the capacity of the train are required in the first few cycles and they cannot be accommodated by early loading, emergency train routes can be used. This case is not considered in the loading problem below. So we assume that the system can accommodate the peak demand caused by disturbances without the need for the emergency routes.

The following variable and parameter are used

ε_{st} is the difference between the actual line-side inventory and β_s

x_{st} is the decision variable which represents the number of delivered bins for station s in the train cycle t.

If there is no early loading, x_{st} and d_{st} will be identical. The model is as follows:

Objective function

$$\min \sum_{s=1}^{N} \sum_{t=1}^{T} |\varepsilon_{st}| \tag{19}$$

Constraints

$$\sum_{s=1}^{N} x_{st} \leqslant K \quad \forall t = 1, \ldots, T \tag{20}$$

$$\sum_{t'=1}^{t} x_{st'} \geqslant \sum_{t'=1}^{t} d_{st'} - \alpha_s \quad \forall t = 1, \ldots, T, s = 1, \ldots, N \tag{21}$$

$$\sum_{t'=1}^{t} x_{st'} - \sum_{t'=1}^{t} d_{st'} + d_{st} + \alpha_s \leqslant M \quad \forall t = 1, \ldots, T, \quad s = 1, \ldots, N \tag{22}$$

$$\sum_{t'=1}^{t} x_{st'} - \sum_{t'=1}^{t} d_{st'} + \alpha_s - \beta_s = \varepsilon_{st} \quad \forall t = 1, \ldots, T, \quad s = 1, \ldots, N \tag{23}$$

$$x_{st} \text{ is positive integer} \quad \forall t = 1, \ldots, T, \quad s = 1, \ldots, N \tag{24}$$

Constraint (20) is to guarantee that the capacity of the train is not exceeded. Constraint (21) guarantees that the accumulated delivered number of bins is at least as the accumulated demand. In the same constraint, the size of line-side inventory at the moment of disturbance is taken into consideration. This inventory is also considered in constraint (22) which guarantees that the MLSI does not exceed the maximum capacity of the area beside stations. The demand of the station is added to the left hand side because the demanded bin(s) will be delivered and stay near the station for at least a moment. So there must be a capacity for that. The value of deviation stated before is defined in constraint (23) to be used in the objective function. To represent the absolute value for the variable ε_{st}, some changes are added to the objective function and constraints, where

$$\varepsilon_{st} = \varepsilon_{st}^{+} - \varepsilon_{st}^{-}$$
$$|\varepsilon_{st}| = \varepsilon_{st}^{+} + \varepsilon_{st}^{-}$$
$$\varepsilon_{st}^{+}, \varepsilon_{st}^{-} \geqslant 0$$

So the objective function is rewritten as:

$$\min \sum_{s=1}^{N} \sum_{t=1}^{T} \varepsilon_{st}^{+} + \varepsilon_{st}^{-} \tag{25}$$

Constraint (23) is rewritten as

$$\sum_{t'=1}^{t} x_{st'} - \sum_{t'=1}^{t} d_{st'} + \alpha_s - \beta_s = \varepsilon_{st}^{+} - \varepsilon_{st}^{-} \quad \forall t = 1, \ldots, T, \quad s = 1, \ldots, N \tag{26}$$

7. Results and analysis

This section concentrates mainly on the effect of using the dynamic control of material flow and compares it to the traditional demand-oriented and kanban systems. In demand-oriented system a complete failure is expected due to the fact that the accuracy of future expectations about demand is deteriorated after disturbances especially in the case of resequencing. So we concentrate more on comparing our system to kanban system.

To compare the system in this study to kanban system in the case of high level of fill rate, Table 5 shows the fill rate for the two systems. The effect of the current line-side inventory was neglected. K values were chosen so that fill rate is greater than or equal to 99%. Actually the difference comes from the use of early loading. Usually early loading is used in demand-oriented system but it is not appropriate in the case of disturbances. So it is used in our approach. Early loading makes the system almost independent from the number of cycles. It is obvious that the system proposed in this study outperforms the demand-oriented and kanban systems.

Eq. (24) is to find δ_{diff} which is the difference between the two systems in terms of fill rate.

$$\delta_{diff} = \text{CDF}(K, D) - \text{CDF}(K, D)^{T} \tag{27}$$

Another direction to investigate the effect of using our approach is to examine the effect of using fixed routes on the underutilization of

Table 5
Effect of early loading on fill rate.

D^a	K^a	Our system $\delta_{EL}{}^a$	Kanban system δ^a		
			$T^*=10$	$T = 15$	$T = 20$
5	11	0.995	0.947	0.921	0.896
6	12	0.991	0.915	0.875	0.838
7	14	0.994	0.944	0.918	0.892
8	15	0.992	0.921	0.883	0.848
9	17	0.995	0.948	0.923	0.899
10	18	0.993	0.930	0.897	0.866
11	19	0.991	0.911	0.869	0.830
12	21	0.994	0.941	0.913	0.885
13	22	0.992	0.926	0.892	0.858
14	23	0.991	0.911	0.869	0.829
15	25	0.994	0.940	0.911	0.883

[a] D is average cell demand during the train cycle time, K is train capacity, δ is fill rate, δ_{EL} is fill rate if early loading is used, and T is number of the train cycles in the shift.

trains. We plot the $(K - D)/D$ value found by Eq. (28) against D. The nominator represents the extra needed capacity higher than the average cell demand. If this value is so large compared to the average cell demand, the underutilization of trains is high. Fig. 3 shows the results for $\delta_{EL} = 99\%$.

$$\frac{K - D}{D} \cong \frac{F^{-1}(\delta_{EL}, D) - D}{D} \tag{28}$$

It is clear from Fig. 3 that using fixed routes for low level of demand (number of bins) affects severely the underutilization of trains. However, when the average demand is high, the level of utilization is enhanced. In the case that there are only some few stations in the cell, there is a probability that MLSI affects the results when MLSI > M, and hence the performance of early loading will not only depends on the accumulated demand of stations but also on the MLSI value. This is because there are only few possibilities for early loaded quantity which may be added to the already large demanded quantity in a certain cycle. However, from the figure above it is not recommended to use fixed routes for the cell with small number of stations because in this case the demand will be low, and the low cell demand increases the underutilization of tugger trains.

The example in scheduling problem shows the possibility of getting different solutions based on the weights of the terms of the objective function. These weights can be set by the decision maker. The main trade-off is between reducing the number of train cycles to decrease the material handling costs, and reducing the probability of early loading and hence reducing the inventory costs. For the loading problem which is necessary if the demand in some train cycles is more than the capacity of the train, it is important to know that the penalty of deviation of the current line side inventory from its ideal level is measured in two dimensions: the quantity (number of bins) and time (number of train cycles). For example, if a deviation of one bin occurs in the first two cycles, the penalty will be 2 not 1.

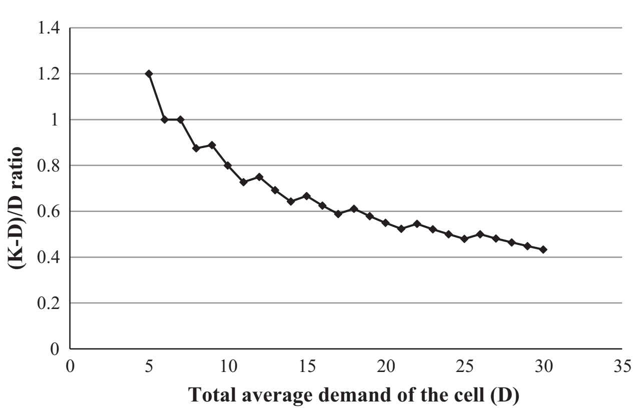


Fig. 3. Effect of fixed routing on underutilization of trains.

8. Conclusion

In this study, an approach for dynamic material flow control is presented. This approach accommodates the effect of assembly line disturbances on the fluctuations of the stations demand for materials and parts. We use a mix between the demand-oriented and e-kanban systems. To fit into this situation, fixed routing is presented taking into account the effect of early loading. Moreover, scheduling and loading problems were investigated taking into consideration the current available amount of inventory beside stations. The objectives considered are minimizing the possibility of early loading, number of train cycles, and the deviations from the ideal safety stock size. These objectives guarantee minimizing inventory holding costs and traffic jam of tugger trains. The presented approach outperforms the kanban system by enhancing the fill rate and that means better utilization of tugger trains capacity. This is especially obvious when the number of train cycles in the shift increases, because the approach combining the e-kanban with the demand–oriented system is independent from the number of train cycles. This was found true using simulation when the fill rate is high. However, the underutilization level of tugger trains increases when cell demand decreases. Future research can focus on finding a suitable approach for routing in the case that cells are with low demand for bins.

The methodology in this study can only be applied when the workstations demand for parts is previously known with a good level of accuracy if there is no any of disturbances in part of the shift. This is usually found in mixed model assembly line. Knowing the demand in advance absorbs the effect of dynamic workstations demand using early loading. Another limitation is the assumption of fixed routes that have been found in some facilities in the practice. However, in other facilities, management decides to use flexible routing in which new routes can be built after the occurrence of the disturbance. The management can apply the fixed routing when workstations demand for bins is relatively high and when the standardization of work is looked for.

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